

HAT-P-36b

R-0934

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230324 · created 2026-07-29T11:45:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460027.72 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.045
Transit depth (Rp/Rs)^2	9.5e-05 +/- 0.0088 [%]
Orbital Inclination (inc)	86.63 +/- 2.83
Airmass coefficient 1 (a1)	0.71 +/- 0.45
Airmass coefficient 2 (a2)	0.37 +/- 0.52
Scatter in the residuals of the lightcurve fit is	58.31 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	3.0
Transit Duration (day)	0.0849 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.001 $\pm$ 0.045 vs 0.126 (deviation 2.72 σ)
Duration fit vs published	0.0849 d vs 0.0925 d (deviation 1.02 σ)
Mid-time O-C	-2.8 min
Depth SNR	0.0
Residual scatter	58.31 %

Light curve

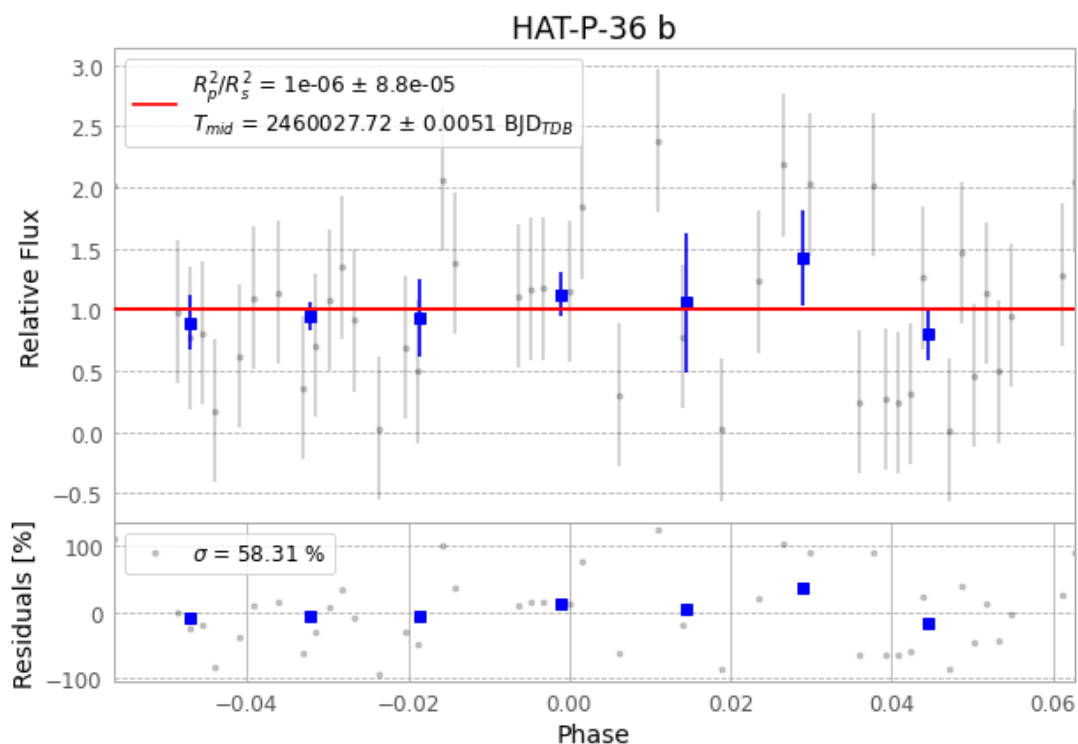


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-03-23.png (SHA-256 1349b50ff8db8551...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [638, 252], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fcb71d79a9b487fd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

78 FITS frames (51 MB), HATP-36230324032115.FITS → HATP-36230324071815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230324.log (346e89e260c4...), FinalLightCurve_HAT-P-36 b_2023-03-23.png (1349b50ff8db...), FinalLightCurve_HAT-P-36 b_2023-03-23.csv (c838c6c9a221...)

Compute resources

Wall time 185.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 11:45Z.